

HealthConnect Week 5 Analytics Report

Data Analytics Track | Prepared by: AnalystLab Africa — Data Analytics Intern

1. Week 4 Transition

Before beginning the Week 5 practical work, this section briefly reviews the Week 4 foundation. It is a transition, not a repeat of the full Week 4 submission.

1. Problem defined. HealthConnect has a high rate of missed appointments (no-shows), which wastes clinical capacity, delays care for other patients, and reduces revenue and staff utilisation. The data analytics track's role is to quantify the pattern and identify its drivers.

2. Week 4 proposed approach. Clean and prepare the data, establish a baseline no-show rate, segment systematically across five candidate KPI dimensions (booking lead time, prior no-show history, reminders, distance, and appointment type), look for interaction effects, and translate findings into recommendations.

3. HealthConnect resources used. The HealthConnect_Appointment_Data dataset (5,000 appointment records, 18 columns) is now imported into Microsoft SQL Server for this week's work, with Power BI planned for the dashboard build in Week 6.

4. Key assumptions/limitations identified in Week 4. Cancelled appointments are excluded from no-show rate calculations (treated as a distinct, proactive outcome); the reminder_channel "None" category is a logical null tied to no reminder being sent, not missing data; small unexplained gaps exist in distance_to_clinic_km and waiting_time_minutes; observed patterns are descriptive/correlational, not causal.

5. Proposed focus for Week 5. Build out the five KPIs fully with supporting visualisations, using SQL Server for data preparation and querying, and produce an initial analytical dashboard ahead of the full Power BI build in Week 6 — the work carried out in this report.

2. Data Preparation

2.1 Dataset Import and Verification

The HealthConnect appointment dataset was imported into Microsoft SQL Server and stored in the HealthConnect_db database. The import was verified by confirming that the dataset contained 5,000 records and 18 columns. Column names, data types, and sample records were also reviewed to confirm that the data was imported correctly.

To preserve the original dataset, a separate working table named HealthConnect_Appointment_Working was created for data preparation and analysis.

2.2 Missing Value Assessment

A missing-value assessment was conducted on the working dataset. The results showed that distance_to_clinic_km contained 90 missing values, while waiting_time_minutes contained 60 missing values.

The reminder_channel column contained no SQL NULL values. However, 1,366 records contained the value None. Further review showed that the reminder indicator uses 0 and 1, where 0 represents no reminder sent and 1 represents a reminder sent. Therefore, None was treated as a valid category representing appointments where no reminder was sent rather than as missing data.

Records with missing distance or waiting-time values were retained because they may still be useful for other analyses. Missing values will only be excluded from calculations involving the affected variables.

Check	Result
Total records	5,000
Total columns	18
Missing distance_to_clinic_km	90
Missing waiting_time_minutes	60
SQL NULLs in reminder_channel	0
None in reminder_channel	1,366
Working table created	Yes

2.3 Duplicate Assessment

Duplicate checks were conducted using appointment_id and a comparison of all 18 variables. The dataset contained 5,000 total records and 5,000 unique appointment IDs, confirming that each appointment had a unique identifier. No completely duplicated records were identified. Therefore, no duplicate records were removed during data preparation.

2.4 Data Consistency and Validity Checks

A data consistency and validity assessment was conducted on the HealthConnect appointment dataset to identify inconsistent, invalid, or logically incorrect values.

Categorical Variables

The categorical variables reviewed included:

- Gender
- Age group
- Appointment type
- Appointment day
- Appointment time
- Reminder sent
- Reminder channel
- Appointment outcome

All categories were found to contain consistent and valid values. No unexpected labels, spelling inconsistencies, or invalid categories were identified.

The reminder_sent variable contained only the values 0 and 1, representing whether a reminder was not sent or was sent. The reminder_channel variable was also logically consistent with this indicator. Records with no reminder sent were recorded as None, while appointments with reminders were associated with valid reminder channels.

Numeric Variables

Numeric variables were assessed to identify invalid or impossible values. The variables reviewed included:

- Age
- Booking lead days
- Previous appointments
- Previous no-shows
- Distance to clinic
- Waiting time

The assessment found no negative values or values outside the expected range. The numeric data was considered valid for further analysis, with the exception of the missing values previously identified in distance_to_clinic_km and waiting_time_minutes.

Logical Consistency Checks

Relationships between relevant variables were also examined to ensure logical consistency. The following checks were completed:

- Previous no-shows did not exceed previous appointments.
- No appointment date occurred before its booking date.
- booking_lead_days matched the calculated difference between booking date and appointment date.
- appointment_day matched the actual day derived from the appointment date.
- Reminder status was consistent with the recorded reminder channel.

All records passed these logical consistency checks.

Final Finding

Overall, the HealthConnect dataset demonstrated good data consistency and validity. No duplicate records, invalid categorical values, invalid numeric values, or logical inconsistencies were identified. The only data quality issues observed were 90 missing values in distance_to_clinic_km and 60 missing values in waiting_time_minutes. These records were retained because the remaining variables may still be useful for other areas of the analysis. Missing values will be handled appropriately when analysing metrics involving the affected variables.

2.5 Data Preparation Summary and Final Readiness Check

The HealthConnect appointment dataset was prepared for exploratory analysis using Microsoft SQL Server. To preserve the original data, the imported dataset was retained without modification, and a separate working table, HealthConnect_Appointment_Working, was created for data preparation and analysis.

The dataset contains 5,000 appointment records and 18 variables. The import was verified by reviewing the dataset structure, column names, data types, and sample records.

A missing value assessment identified 90 missing values in distance_to_clinic_km and 60 missing values in waiting_time_minutes. The reminder_channel column contained 1,366 records labelled None. This was investigated alongside the reminder_sent variable and was confirmed to represent appointments where no reminder was sent. Therefore, None was treated as a valid category rather than a missing value.

Duplicate checks confirmed that all 5,000 appointment records had unique appointment IDs, and no completely duplicated records were identified.

Further data consistency and validity checks confirmed that categorical variables contained consistent values, numeric variables contained no invalid values, and the relationships between relevant variables were logically correct. For example, previous no-shows did not exceed previous appointments, appointment dates did not occur before booking dates, and reminder status was consistent with reminder channel information.

Final Data Preparation Decision

Based on the checks completed, the dataset is considered suitable for exploratory data analysis. The missing values in distance_to_clinic_km and waiting_time_minutes will not result in the removal of entire records. Instead, they will be considered during analyses involving those specific variables. The original dataset remains unchanged, while all subsequent analysis will be conducted using the working table.

Data Preparation Readiness Check

Assessment Area	Result	Status
Import verification	5,000 records and 18 columns confirmed	Complete
Working table	Created successfully	Complete
Missing values	90 distance and 60 waiting time values identified	Documented
Reminder channel	None confirmed as a valid category	Verified
Duplicate appointment IDs	None found	Complete
Completely duplicated records	None found	Complete
Categorical consistency	No inconsistencies found	Complete
Numeric validity	No invalid values found	Complete
Logical consistency	All checks passed	Complete
Dataset readiness	Ready for EDA	Ready

3. Exploratory Data Analysis

Exploratory analysis was conducted on the working table to investigate appointment attendance and no-show patterns across appointment characteristics, patient characteristics, appointment history, reminders, waiting time, distance, and cancellations.

3.1 Overall Attendance Picture

HealthConnect recorded a 48.46% no-show rate, with 2,423 appointments missed out of 5,000. Attended appointments accounted for 46.28%, while 5.26% were cancelled. This indicates that missed appointments represent a significant operational challenge for HealthConnect and may result in inefficient use of healthcare resources.

3.2 Appointment Type

Follow-up appointments recorded the highest no-show rate at 51.23%, followed by diagnostic tests at 49.75%, specialist consultations at 47.44%, and general consultations at 46.64%. This suggests that appointment type may influence attendance behaviour, with follow-up appointments requiring particular attention.

3.3 Age Group

No-show rates varied across age groups. The 55–64 age group recorded the highest rate at 50.75%, closely followed by 25–34 at 50.70% and 18–24 at 50.18%. Patients aged 65 and above had the lowest rate at 45.12%. Although the differences are not extremely large, age may contribute to variations in appointment attendance.

3.4 Appointment Reminders

Patients who did not receive reminders recorded a 51.39% no-show rate, compared with 47.36% among patients who received reminders. This suggests that reminders may help improve appointment attendance and reduce missed appointments. This is an association, not proof that reminders directly caused the improvement.

3.5 Waiting Time

No-show rates generally increased as waiting time increased. Shorter waiting periods were associated with lower no-show rates, while longer waiting periods recorded higher rates. This suggests that longer waiting times may increase the likelihood of patients missing their appointments.

3.6 Booking Lead Time

Booking lead time showed one of the clearest patterns in the analysis. Patients who booked 0–7 days in advance had the lowest no-show rate at 27.81%, rising to 33.55% for 8–14 days, 43.21% for 15–30 days, and 60.49% for 31 or more days. This suggests that appointments scheduled far in advance may be more vulnerable to being forgotten, changed, or missed.

3.7 Distance to the Clinic

Patients living more than 20 km from the clinic had the highest no-show rate at 56.35%, compared with 46.40% among patients living less than 5 km away. This suggests that greater distance from the clinic may increase the likelihood of patients missing their appointments.

3.8 Previous No-Show Behaviour

A strong relationship was observed between previous missed appointments and future no-shows. Patients with no previous no-shows had a 43.51% no-show rate, rising to 53.49% with one previous

no-show, 59.36% with two, and 67.95% with three. This indicates that previous attendance behaviour may be one of the strongest indicators of future no-show risk. The groups with four and five previous no-shows were very small and should be interpreted cautiously.

3.9 Appointment Day

No-show rates varied by appointment day. Sunday had the highest no-show rate at 50.47%, followed by Wednesday at 49.66% and Monday at 49.64%. Friday had the lowest rate at 46.57%. The differences were relatively small, suggesting appointment day has some influence but is not as strong a factor as booking lead time or previous no-show history.

3.10 Appointment Time

The appointment_time column contains the categories Morning, Afternoon, and Evening. This comparison is still pending: an earlier attempt to convert these category labels into a time value incorrectly returned all 5,000 appointments as "Unknown", and that result has been excluded from this report. A category-level comparison of no-show rates across morning, afternoon, and evening will be completed in a follow-up pass.

4. KPI Development and Results

Five KPIs were carried forward from the Week 4 candidate list, each defined, calculated on the SQL working table, linked to a business question, and interpreted below.

KPI 1 — Overall No-Show Rate

Definition: $\text{No-Shows} \div (\text{Attended} + \text{No-Show})$, excluding cancelled appointments.

Why it matters: This is the single headline number HealthConnect needs to track — it quantifies the scale of lost clinical capacity and is the baseline every intervention should be measured against.

Linked business question: Q1 — what is the overall scale of the no-show problem?

Metric	Result
Overall No-Show Rate	48.46%
Total No-Shows	2,423 of 5,000

Interpretation: Nearly 1 in 2 appointments (48.46%) ends in a no-show. This confirms the business problem is significant enough to warrant active intervention, not just monitoring.

KPI 2 — No-Show Rate by Booking Lead Time Band

Definition: No-show rate segmented into 0–7 / 8–14 / 15–30 / 31+ day bands between booking and appointment.

Why it matters: If lead time drives no-shows, HealthConnect could adjust scheduling policy — e.g. cap advance booking windows or add extra reminder touchpoints for far-out bookings.

Linked business question: Q2 — does booking lead time affect no-show likelihood?

Lead Time Band	No-Show Rate
0–7 days	27.81%
8–14 days	33.55%
15–30 days	43.21%
31+ days	60.49%

Interpretation: No-show rate more than doubles across the range. This is the clearest pattern in the dataset and the strongest actionable lever: appointments booked far in advance need stronger follow-up.

KPI 3 — No-Show Rate by Prior No-Show History

Definition: No-show rate segmented by the patient's previous_no_shows count.

Why it matters: Identifies whether past behaviour predicts future behaviour, letting HealthConnect flag high-risk patients for targeted outreach before they no-show again.

Linked business question: Q3 — is prior no-show history predictive of a future no-show?

Prior No-Shows	No-Show Rate
0	43.51%
1	53.49%
2	59.36%
3	67.95%
4 and 5	Small sample — interpret with caution

Interpretation: The no-show rate rises steadily with prior no-show count, from 43.51% with no history to 67.95% with three prior no-shows. This is a strong, actionable signal for risk-based targeting.

KPI 4 — Reminder Effectiveness

Definition: No-show rate compared across reminder_sent (reminder sent vs. not sent).

Why it matters: Directly measures whether the reminder programme works, informing whether and where to expand reminder coverage.

Linked business question: Q4 — does sending a reminder reduce no-show rates?

Reminder Status	No-Show Rate
No reminder sent	51.39%
Reminder sent	47.36%

Interpretation: Reminders reduce the no-show rate by just over 4 percentage points (51.39% → 47.36%). This supports continuing and expanding the reminder program while recognising this is an association rather than proven causation.

KPI 5 — No-Show Rate by Distance Band

Definition: No-show rate segmented by distance_to_clinic_km.

Why it matters: Tests whether physical access to the clinic is a barrier, pointing towards different interventions (e.g. transport support, telehealth) than a reminder-based fix.

Linked business question: Q5 — does distance from the clinic relate to no-show risk?

Distance Band	No-Show Rate
Less than 5 km	46.40%
More than 20 km	56.35%

Interpretation: The no-show rate rises by roughly 10 points between the closest and furthest patients. Distance is a contributing factor, not the primary driver, but worth flagging for patients furthest from the clinic.

Booking lead time and prior no-show history stand out as the strongest levers in the dashboard, with reminders, appointment type, and distance as secondary but still actionable factors.

5. Key Business Insights

1. HealthConnect has a significant no-show problem.

Out of 5,000 appointments, 2,423 resulted in no-shows, representing a 48.46% no-show rate. This means almost one out of every two scheduled appointments was missed, which can affect clinic efficiency, staff planning, and the use of available appointment slots.

2. Follow-up appointments recorded the highest no-show rate

Among appointment types, follow-up appointments had the highest no-show rate at 51.23%. This suggests that patients may place less urgency on follow-up visits compared to other appointment types.

3. Reminders were associated with better attendance.

Patients who received reminders had a lower no-show rate of 47.36%, compared with 51.39% for patients who did not receive reminders. This suggests that reminders may play a useful role in reducing missed appointments.

4. Longer booking lead times were strongly associated with higher no-show rates.

Appointments booked closer to the appointment date had much lower no-show rates. The rate increased from 27.81% for appointments booked 0–7 days in advance to 60.49% for appointments booked 31 or more days in advance. This was one of the clearest patterns identified in the analysis.

5. Previous no-show history is an important risk indicator.

Patients with previous missed appointments were more likely to miss future appointments. The no-show rate increased as the number of previous no-shows increased. This suggests that HealthConnect can use previous attendance behaviour to identify patients who may be at higher risk.

6. Longer waiting periods may contribute to missed appointments.

The analysis showed a general increase in no-show rates as waiting time increased. Patients may become less committed, forget their appointments, or experience changes in their schedules during longer waiting periods.

7. Distance may create an attendance barrier.

Patients living more than 20 km from the clinic recorded a 56.35% no-show rate, compared with 46.40% among patients living less than 5 km away. This suggests that travel distance may be a barrier to appointment attendance.

8. Age and appointment day showed smaller variations.

No-show rates varied across age groups and appointment days, but these differences were less pronounced than the patterns observed for booking lead time, previous no-show history, waiting time, reminders, and distance.

7. Business Recommendations

Recommendation 1: Implement a targeted reminder system

HealthConnect should continue using appointment reminders and improve the process by giving additional attention to patients with a higher risk of missing appointments.

For example:

- Send an initial confirmation after booking
- Send a reminder a few days before the appointment
- Send a final reminder closer to the appointment time
- Allow patients to easily confirm, cancel, or reschedule

This can help reduce forgotten appointments and allow cancelled slots to be reassigned.

Recommendation 2: Flag patients with previous no-shows

Patients with a history of missed appointments should be identified as higher-risk patients. HealthConnect can create a simple risk flag based on:

- Number of previous no-shows
- Booking lead time
- Waiting time
- Distance to the clinic

Higher-risk patients can then receive additional reminders or follow-up communication.

Recommendation 3: Reduce long booking lead times where possible

Appointments scheduled 31 or more days in advance recorded the highest no-show rate. Where possible, HealthConnect should:

- Avoid scheduling appointments too far in advance
- Use shorter booking windows
- Reconfirm appointments booked several weeks earlier
- Send stronger reminder sequences for long-lead appointments

Recommendation 4: Review long waiting periods

HealthConnect should monitor appointments with long waiting periods and identify opportunities to reduce delays. Where long waiting periods cannot be avoided, additional communication should be provided to keep patients informed and engaged.

Recommendation 5: Improve support for patients travelling long distances

Patients living farther from the clinic may face transportation and travel challenges. HealthConnect could consider:

- Earlier appointment confirmation
- Flexible scheduling where possible
- Virtual consultations for suitable appointment types
- Prioritised communication for patients travelling longer distances

Recommendation 6: Create a no-show monitoring dashboard

HealthConnect should regularly monitor appointment performance using key metrics such as:

- Total appointments
- No-show rate
- Attendance rate
- No-show rate by appointment type
- No-show rate by age group
- Reminder status
- Booking lead time
- Previous no-show history
- Distance to clinic
- Appointment day and time

This is where the Power BI dashboard becomes valuable. It can help HealthConnect track these patterns continuously rather than performing a one-time analysis.

8. Limitations

- All relationships reported here are descriptive/correlational, not causal — for example, the reminder effect is a genuine association, but a controlled comparison would be needed before treating it as a proven causal impact.
- The prior-no-show groups with 4 and 5 previous no-shows are based on very small sample sizes and should not be treated as reliable rates; the 0–3 gradient is based on much larger, more dependable samples.

- The waiting_time_minutes variable's exact definition should be confirmed with the project team — this analysis treats it as a general waiting-period measure, and the strength of its relationship with no-shows should be re-checked once the definition is verified, since it may overlap conceptually with booking lead time.
- distance_to_clinic_km and waiting_time_minutes carry small (1–2%) gaps that were excluded only from calculations directly involving those variables, not from the dataset as a whole.
- The appointment_time (morning/afternoon/evening) comparison is still pending, following a data-conversion issue that produced an invalid “Unknown” result; this will be corrected and re-run before the Week 6 dashboard build.
- Age and appointment-day differences were modest and should be treated as secondary factors rather than primary drivers.
- This week's KPI and dashboard work was completed in SQL Server and a static chart; the full interactive Power BI version, planned for Week 6, may reveal additional patterns once cross-filtering across variables is possible.

9. Week 5 Project Summary

- 1. What I planned to accomplish.** Build out the five Week 4 candidate KPIs fully with supporting visualisations, using SQL Server for data preparation, and begin scoping the Power BI dashboard structure for the reporting phase.
- 2. What I actually completed.** Full data preparation in Microsoft SQL Server on a dedicated working table (source table untouched); a missing-value, duplicate, and consistency assessment; exploratory analysis across ten dimensions of the appointment data; all five KPIs calculated and interpreted; an initial six-panel analytical dashboard; and eight business insights with recommendations.
- 3. Key findings/development outcomes.** Booking lead time and prior no-show history are the strongest drivers of no-shows; reminders show a real, measurable effect; distance and appointment type are secondary but relevant factors; age and appointment day show smaller variation; the appointment-time comparison remains outstanding.
- 4. Major challenges encountered.** An initial attempt to compare no-show rates by appointment_time produced an invalid “Unknown” result due to a data-type conversion issue; this was identified and excluded from the report rather than reported as a finding and will be corrected in a follow-up pass.
- 5. Important decisions made and why.** Cancelled appointments were kept out of the no-show rate denominator throughout, consistent with the Week 4 assumption; the reminder_channel “None” category was confirmed and treated as a valid “no reminder” category rather than missing data after cross-checking against reminder_sent; missing values in distance and waiting time were retained in the dataset and excluded only from calculations directly involving those fields.
- 6. Changes to the Week 4 approach.** Data preparation and KPI calculation moved from a spreadsheet-style exploration to Microsoft SQL Server, using a dedicated working table — a more scalable approach consistent with the track's stated tools. No candidate KPIs were dropped; all five were carried forward.
- 7. Cross-track collaboration completed.** None yet this week; the KPI definitions and dashboard groundwork here are intended as the handoff point for the BI/reporting track's Power BI build in Week 6.

8. Remaining work. Correct and complete the appointment_time comparison; translate the five KPIs definitions into a reusable set of SQL views; rebuild the dashboard as an interactive Power BI report; validate the reminder and waiting-time effects with more rigors comparisons before they inform budget or policy decisions.

9. Proposed focus for Week 6. Migrate the KPI logic into SQL views, resolve the appointment_time analysis, and build the first interactive Power BI dashboard version of the panels shown here, ready for stakeholder review.